Unit 8: Point Estimation

8.1: Introduction
8.2 – 8.4: Bias and Error
8.3: Common Unbiased Point Estimators
9.2: Relative Efficiency
9.3: Consistency
9.6: Method of Moments Estimation
9.7: Maximum Likelihood Estimation
9.4: Sufficiency
9.5: Rao-Blackwell Theorem and MVUE

Optional
9.8: Large Sample Properties of MLE's (optional)

Introduction

Recall Definition 7.1 : Statistic

⇒ A **statistic** is any function of random variables collected in a sample, and known constants. Statistics are random variables.

Definition 8.1 : Estimators

⇒ An **estimator** is a rule or formula that defines how to calculate an estimate for a population parameter based on observations collected in a sample. Estimators are statistics.

⇒ A single value will be called a **Point Estimator.**

⇒ An interval of values will be called an **Interval Estimator**.

8.2 – 8.4: Bias and Error

In general: $\theta =$ target parameter, $\hat{\theta}$ or $\hat{\theta}_n =$ point estimator

Definition 8.2 : Unbiased Estimator

$\Rightarrow \hat{\theta}$ is said to be an **unbiased estimator** for θ if $E\left[\hat{\theta}\right] = \theta.$

If $E\left[\hat{\theta}\right] \neq \theta,$ then $\hat{\theta}$ is said to be **biased** for $\theta.$

Definition 8.3 : Bias

$\Rightarrow$ The **bias** of a point estimator is $B\left[\hat{\theta}\right] = E\left[\hat{\theta}\right] - \theta.$

Example 1:

Suppose $\{Y_1, \ldots, Y_n\}$ is a random sample of independent and identically distributed random variables with mean μ_Y and variance σ_Y^2. Consider the following estimators for the parameter μ_Y :

$$\hat{\mu}_1 = \overline{Y}, \quad \hat{\mu}_2 = \frac{Y_1 + Y_2}{3}, \quad \hat{\mu}_3 = \frac{5Y_3 + Y_7 - 2Y_n}{3}, \quad \hat{\mu}_4 = \frac{2}{n(n+1)} \sum_{i=1}^{n} iY_i.$$

a) Which estimators are unbiased? Which are biased? Calculate the bias for each estimator.

General: Sample Means are Unbiased for Population Means

Recall: For a random variable Y, the k-th **raw moment** is $\mu_k' = E\left[Y^k\right]$.

Definition: The k-th **sample raw moment**, based on a random sample of size n is

$$m_k' = \frac{1}{n}\sum_{i=1}^{n} Y_i^k.$$

Consider estimating the raw moments using the sample raw moments:

That is, $\theta = \mu_k'$ and $\hat{\theta} = m_k'$.

$$\Rightarrow E\left[\hat{\theta}\right] = E\left[m_k'\right] = E\left[\frac{1}{n}\sum_{i=1}^{n} Y_i^k\right] = \frac{1}{n} nE\left[Y^k\right] = \mu_k' = \theta.$$

Example 2:

Suppose $\{Y_1, \ldots, Y_n\}$ is a random sample of independent and identically distributed random variables with mean μ_Y and variance σ_Y^2. Consider the following estimators for the parameter σ_Y^2 :

- $\hat{\sigma}_1^2 = c_1 \sum_{i=1}^{n} (Y_i - \mu_Y)^2$, where μ_Y is assumed to known.
- $\hat{\sigma}_2^2 = c_2 \sum_{i=1}^{n} (Y_i - \bar{Y})^2$.

a) What are the constants, c_1 and c_2, so that these are unbiased estimators for σ_Y^2 ?

Definition 8.4 : Mean Square for Error

$\Rightarrow$ The **mean square for error** (MSE) of a point estimator $\hat{\theta}$ is

$$MSE\left[\hat{\theta}\right] = E\left[\left(\hat{\theta} - \theta\right)^2\right].$$

Notes:

- $MSE\left[\hat{\theta}\right] = V\left[\hat{\theta}\right] + \left(B\left[\hat{\theta}\right]\right)^2.$
- If $\hat{\theta}$ is an unbiased estimator then $MSE\left[\hat{\theta}\right] = V\left[\hat{\theta}\right].$

Definition 9.1 : Relative Efficiency

Consider two unbiased estimators, $\hat{\theta}_1$ and $\hat{\theta}_2$, for θ.

$\Rightarrow$ The **efficiency** of $\hat{\theta}_1$ relative to $\hat{\theta}_2$ is $eff\left(\hat{\theta}_1, \hat{\theta}_2\right) = \dfrac{V\left(\hat{\theta}_2\right)}{V\left(\hat{\theta}_1\right)}.$

- If $eff\left(\hat{\theta}_1, \hat{\theta}_2\right) > 1,$ then $\hat{\theta}_1$ is more efficient (better).
- If $eff\left(\hat{\theta}_1, \hat{\theta}_2\right) < 1,$ then $\hat{\theta}_2$ is more efficient.

Example 1:

Suppose $\{Y_1, \ldots, Y_n\}$ is a random sample of independent and identically distributed random variables with mean μ_Y and variance σ_Y^2. Consider the following estimators for the parameter μ_Y :

$$\hat{\mu}_1 = \overline{Y}, \quad \hat{\mu}_2 = \frac{Y_1 + Y_2}{3}, \quad \hat{\mu}_3 = \frac{5Y_3 + Y_7 - 2Y_n}{3}, \quad \hat{\mu}_4 = \frac{2}{n(n+1)} \sum_{i=1}^{n} iY_i.$$

b) Find the MSE for each estimator. Which unbiased estimator is the most efficient?

Example 2:

Suppose $\{Y_1, \ldots, Y_n\}$ is a random sample of independent and identically distributed ***Normal*** random variables with mean μ_Y and variance σ_Y^2. Consider the following unbiased estimators for the parameter σ_Y^2:

- $\hat{\sigma}_1^2 = \frac{1}{n}\sum_{i=1}^{n}(Y_i - \mu_Y)^2$, where μ_Y is assumed to known.
- $\hat{\sigma}_2^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$.

Calculate the efficiency of $\hat{\sigma}_1^2$ relative to $\hat{\sigma}_2^2$.